qdrant / qdrant Public

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generall trigger CI ✓ eabee37 · last month

.config	Decrease retry count on flaky test ...	11 months ago
.github	ci: Pin all gh actions to commit SH...	last month
config	add tracing ID into audit logging (#...	last month
docs	Feature/strict mode search max ba...	last month
lib	bump version to v1.17.1 (#8517)	last month
openapi	Implement Per collection metrics f...	last month
pkg/appimage	build: AppImage release (#3343)	2 years ago
src	Feature/strict mode search max ba...	last month
tests	Feature/strict mode search max ba...	last month
tools	bump version to v1.17.1 (#8517)	last month
.dockerignore	disk cache hygiene (#6323)	last year
.gitattributes	fix(core): optimize OCI Layers (#70...	8 months ago
.gitignore	Chores (#8370)	last month
.rusty-hook.toml	Add import formatting rules (#820)	4 years ago
Cargo.lock	bump version to v1.17.1 (#8517)	last month
Cargo.toml	bump version to v1.17.1 (#8517)	last month
Dockerfile	build(deps): bump lukemathwalker...	last month
LICENSE	Create LICENSE	5 years ago
README.md	chore(docs): fix llamaindex link in r...	last month
clippy.toml	Streaming snapshot unpacking (#...	2 months ago
rustfmt.toml	Add import formatting rules (#820)	4 years ago
shell.nix	Test edge examples on CI (#8318)	last month

README | Code of conduct | Contributing | Apache-2.0 license

Qdrant

Vector Search Engine for the next generation of AI applications

build passing | Docs OpenAPI 3.0 | license Apache-2.0 | chat 767 online | Roadmap 2025 | Qdrant Cloud

Qdrant (read: *quadrant*) is a vector similarity search engine and vector database. It provides a production-ready service with a convenient API to store, search, and manage points—vectors with an additional payload Qdrant is tailored to extended filtering support. It makes it useful for all sorts of neural-network or semantic-based matching, faceted search, and other applications.

Qdrant is written in Rust 🦀, which makes it fast and reliable even under high load. See benchmarks.

With Qdrant, embeddings or neural network encoders can be turned into full-fledged applications for matching, searching, recommending, and much more!

Qdrant is also available as a fully managed Qdrant Cloud ☁️ including a **free tier**.

Quick Start • Client Libraries • Demo Projects • Integrations • Contact

Getting Started

Python

```
pip install qdrant-client
```

The python client offers a convenient way to start with Qdrant locally:

```
from qdrant_client import QdrantClient
qdrant = QdrantClient(":memory:") # Create in-memory Qdrant instance, for testing, CI/CD
# OR
client = QdrantClient(path="path/to/db")  # Persists changes to disk, fast prototyping
```

Client-Server

To experience the full power of Qdrant locally, run the container with this command:

```
docker run -p 6333:6333 qdrant/qdrant
```

> Caution

> Starts an insecure deployment without authentication open to all network interfaces. Please refer to secure your instance.

Now you can connect to this with any client, including Python:

```
qdrant = QdrantClient("http://localhost:6333") # Connect to existing Qdrant instance
```

Before deploying Qdrant to production, be sure to read our installation and security guides.

Clients

Qdrant offers the following client libraries to help you integrate it into your application stack with ease:

- Official:
  - Go client
  - Rust client
  - JavaScript/TypeScript client
  - Python client
  - .NET/C# client
  - Java client
- Community:
  - Elixir
  - PHP
  - Ruby
  - Java

Where do I go from here?

- Quick Start Guide
- End to End Colab Notebook demo with SentenceBERT and Qdrant
- Detailed Documentation are great starting points
- Step-by-Step Tutorial to create your first neural network project with Qdrant

Demo Projects

Run on Replit

Discover Semantic Text Search 🔍

Unlock the power of semantic embeddings with Qdrant, transcending keyword-based search to find meaningful connections in short texts. Deploy a neural search in minutes using a pre-trained neural network, and experience the future of text search. Try it online!

Explore Similar Image Search - Food Discovery 🍕

There's more to discovery than text search, especially when it comes to food. People often choose meals based on appearance rather than descriptions and ingredients. Let Qdrant help your users find their next delicious meal using visual search, even if they don't know the dish's name. Check it out!

Master Extreme Classification - E-commerce Product Categorization 📺

Enter the cutting-edge realm of extreme classification, an emerging machine learning field tackling multi-class and multi-label problems with millions of labels. Harness the potential of similarity learning models, and see how a pre-trained transformer model and Qdrant can revolutionize e-commerce product categorization. Play with it online!

▶ More solutions

API

REST

Online OpenAPI 3.0 documentation is available here. OpenAPI makes it easy to generate a client for virtually any framework or programming language.

You can also download raw OpenAPI definitions.

gRPC

For faster production-tier searches, Qdrant also provides a gRPC interface. You can find gRPC documentation here.

Features

Filtering and Payload

Qdrant can attach any JSON payloads to vectors, allowing for both the storage and filtering of data based on the values in these payloads. Payload supports a wide range of data types and query conditions, including keyword matching, full-text filtering, numerical ranges, geo-locations, and more.

Filtering conditions can be combined in various ways, including `should`, `must`, and `must_not` clauses, ensuring that you can implement any desired business logic on top of similarity matching.

Hybrid Search with Sparse Vectors

To address the limitations of vector embeddings when searching for specific keywords, Qdrant introduces support for sparse vectors in addition to the regular dense ones.

Sparse vectors can be viewed as an generalization of BM25 or TF-IDF ranking. They enable you to harness the capabilities of transformer-based neural networks to weigh individual tokens effectively.

Vector Quantization and On-Disk Storage

Qdrant provides multiple options to make vector search cheaper and more resource-efficient. Built-in vector quantization reduces RAM usage by up to 97% and dynamically manages the trade-off between search speed and precision.

Distributed Deployment

Qdrant offers comprehensive horizontal scaling support through two key mechanisms:

1. Size expansion via sharding and throughput enhancement via replication
2. Zero-downtime rolling updates and seamless dynamic scaling of the collections

Highlighted Features

- **Query Planning and Payload Indexes** - leverages stored payload information to optimize query execution strategy.
- **SIMD Hardware Acceleration** - utilizes modern CPU x86-x64 and Neon architectures to deliver better performance.
- **Async I/O** - uses `io_uring` to maximize disk throughput utilization even on a network-attached storage.
- **Write-Ahead Logging** - ensures data persistence with update confirmation, even during power outages.

Integrations

Examples and/or documentation of Qdrant integrations:

- Cohere (blogpost on building a QA app with Cohere and Qdrant) - Use Cohere embeddings with Qdrant
- DocArray - Use Qdrant as a document store in DocArray
- Haystack - Use Qdrant as a document store with Haystack (blogpost).
- LangChain (blogpost) - Use Qdrant as a memory backend for LangChain.
- LlamaIndex - Use Qdrant as a Vector Store with LlamaIndex.
- OpenAI - ChatGPT retrieval plugin - Use Qdrant as a memory backend for ChatGPT
- Microsoft Semantic Kernel - Use Qdrant as persistent memory with Semantic Kernel

Contacts

- Have questions? Join our Discord channel or mention @qdrant_engine on Twitter
- Want to stay in touch with latest releases? Subscribe to our Newsletters
- Looking for a managed cloud? Check pricing, need something personalised? We're at info@qdrant.tech

License

Releases 111

v1.17.1 Latest
on Mar 27

+ 110 releases

Packages 2

qdrant/qdrant

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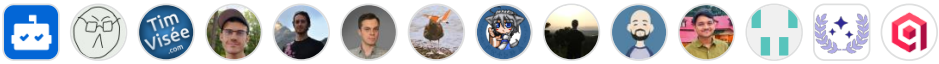
Contributors 163



+ 149 contributors

Languages

Rust 87.4% Python 11.4% Shell 0.7% C 0.3% Nix 0.1% Dockerfile 0.1%